Synopsis

Project Title

Air Quality Prediction using Machine Learning

Problem ID: 15

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Problem Description

Air pollution is a serious environmental issue that affects health and daily life. This project develops a machine learning system to predict air quality levels using pollutant concentration and weather data such as PM2.5, PM10, temperature, humidity, and wind speed. The system classifies air quality into categories like Good, Moderate, Poor, etc., helping in awareness and decision-making.

Steps of Implementation

Data Collection	Collect dataset containing pollutant and meteorological data.
Data Preprocessing	Handle missing values, clean dataset, normalize values.
Feature Engineering	Extract features like date-time components.
Feature Selection	Select important features to improve performance.
Model Selection	Use Logistic Regression, KNN, and SVM.
Training & Validation	Split dataset and train models.
Evaluation	Evaluate using accuracy, precision, recall, F1-score.
Final Model	Select best model for prediction.

Expected Modules/Libraries

Scikit-learn: ML algorithms
NumPy: Numerical computations
Pandas: Data handling
Matplotlib/Seaborn: Visualization

Evaluation Metrics

Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC Curve

1. Accuracy:
Accuracy measures the overall correctness of the model. It tells us how many predictions were correct out of the total predictions made.
Formula:
Accuracy = (True Positive + True Negative) / Total Predictions
Example: If the model correctly predicts 90 out of 100 cases, the accuracy is 90%.
2. Precision:
Precision measures how many of the predicted positive cases are actually correct. It focuses on the quality of positive predictions.
Formula:
Precision = True Positive / (True Positive + False Positive)
Use Case: Important when false positives are costly.
3. Recall (Sensitivity):
Recall measures how many actual positive cases were correctly identified by the model.
Formula:
Recall = True Positive / (True Positive + False Negative)
Use Case: Important when missing a positive case is dangerous (e.g., pollution alerts).
4. F1-Score:
F1-score is the harmonic mean of Precision and Recall. It balances both metrics and is useful when you need a trade-off between them.
Formula:
F1 Score = 2 × (Precision × Recall) / (Precision + Recall)
5. Confusion Matrix:
A confusion matrix is a table used to evaluate the performance of a classification model. It shows:
True Positive (TP): Correctly predicted positive cases
True Negative (TN): Correctly predicted negative cases
False Positive (FP): Incorrectly predicted positive cases
False Negative (FN): Incorrectly predicted negative cases
6. ROC Curve and AUC:
The ROC (Receiver Operating Characteristic) curve shows the relationship between True Positive Rate and False Positive Rate.
AUC (Area Under the Curve) measures the overall performance of the model.
AUC closer to 1 → Better model
AUC closer to 0.5 → Poor model

Platform Required

Jupyter Notebook / Python IDE

Books and Link Sources

1. Introduction to Machine Learning - Ethem Alpaydin
2. Pattern Recognition and Machine Learning - Christopher Bishop
3. Scikit-learn Documentation